Next-Day Air Quality Forecasting and Health Alerts for Arnavutköy, Istanbul

Project Proposal (Concept Note) — Assignment 1
MTH4800 Innovation Campus AI Applications · Capstone: AI for the SDGs · Group 9
GitHub repository: github.com/maliksonmez/arnavutkoy-air-quality-forecast

1. Project Name

Next-Day Air Quality Forecasting and Health Alerts for Arnavutköy, Istanbul

2. Overview

Air pollution is a major urban health risk, especially for children, older adults and people with asthma or heart disease. Monitoring portals report what has already been measured, whereas decisions that limit exposure (outdoor exercise, a child's park visit, carrying medication) are made a day in advance. We therefore propose a **station-level next-day forecast** linked to health guidance.

> *"Our tool helps residents of Arnavutköy — especially sensitive groups such as children, older adults and people with asthma or heart disease — plan their next day by forecasting tomorrow's PM10, PM2.5, SO2 and NO2 levels and warning them when WHO guideline levels are likely to be exceeded."*

- **SDG link: SDG 11.6** (reduce the environmental impact of cities, paying special attention to air quality; indicator 11.6.2: urban PM2.5/PM10 levels) is our primary target, and **SDG 3.9** (reduce illnesses and deaths from air pollution; indicator 3.9.1) is secondary [7].
- **Target users:** residents of Arnavutköy/Istanbul and sensitive groups; local decision-makers as secondary users.
- **How we will know it works:** (1) lower forecast error (MAE) than the simple "tomorrow = today" baseline on a held-out test period; (2) the share of days above WHO guideline levels that the tool correctly flags.

3. Background

The WHO 2021 Air Quality Guidelines recommend 24-hour mean levels of **15 µg/m³ for PM2.5, 45 for PM10, 25 for NO2 and 40 for SO2** [1]. In our data, the daily mean exceeded these levels on **36%** of valid days for PM2.5 and NO2 and **13%** for PM10 (SO2: none), which shows that the problem is real and frequent at this station. Istanbul studies show that traffic is a dominant urban source [3], that black carbon follows NO2 with rush-hour peaks [4], and that meteorology strongly affects particle levels [5]. Recent ML work starts from simple baselines, uses lagged pollutant values as key inputs and explains predictions with SHAP [2]; we follow this approach for a single Istanbul station, with a time-ordered evaluation.

4. Key Objectives

4.1 Research Questions

1. Can a model trained on the station's own history forecast next-day daily concentrations more accurately than persistence ("tomorrow = today")?
2. Which inputs — pollutant lags, calendar effects (weekday), meteorology — improve the forecast most?
3. How reliably can the tool warn of days above WHO guideline levels, and where does it fail?

4.2 Key Steps and User Flow

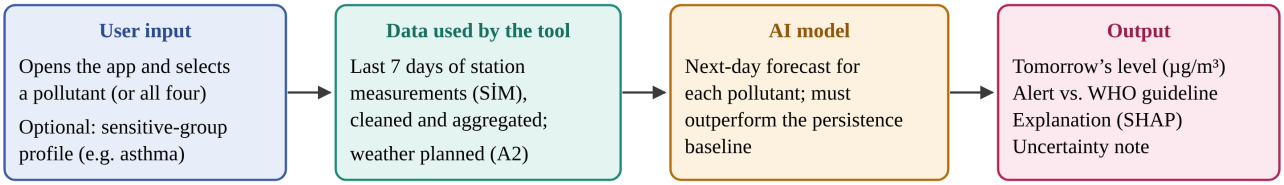


Figure 1. What the user provides and what the tool produces. The full development pipeline is documented in the repository (*docs/flowchart.png*).

Key steps. A1: problem, data, EDA and a first experiment (completed). A2: improved model and a simple end-to-end tool. A3: testing with user profiles, ethical assessment and the final application.

5. Methods and Workflow

5.1 Dataset

Hourly measurements from the **Ministry of Environment, Urbanisation and Climate Change, Continuous Monitoring Centre (SİM)**, İstanbul–Arnavutköy station: **16.09.2023–16.09.2026, 26,305 hourly records, PM10, PM2.5, SO2 and NO2 (µg/m³)**, publicly available [6]. It measures the pollutants behind SDG indicator 11.6.2 and the WHO guidelines, and three years of hourly data capture daily, weekly and seasonal patterns.

5.2 Data Cleaning, Preprocessing and Exploratory Analysis

We followed the course EDA workflow (shape and types → missing and duplicate values → descriptive statistics → distributions → outliers → time series → correlation):

- **Missingness:** no duplicate or skipped hours; 3.9–5.8% missing per pollutant (longest outage 1–9 September 2025). Long gaps were **not** filled, to avoid inventing pollution patterns.
- **Outliers:** distributions are strongly right-skewed (PM10 mean 27.3 vs. median 21.9), so Z-score/IQR rules would flag real episodes; we used context checks instead. Seven identical "985" readings (next-highest values 338 and 186) were treated as likely measurement errors and set to missing in the analysis copy; a sensitivity check showed no change in medians or correlations. Other high values were kept. PM2.5 > PM10 in 12.6% of hours (median difference 2.6 µg/m³) is reported as a quality issue.
- **Aggregation:** a day is valid if it has ≥ 18 hourly values; we use the **daily median**, which is robust to single-hour spikes.
- **Findings:** NO2 peaks at 07–09 and 20–23 (a traffic signature), PM10 is highest in the evening, and Sundays are lower for PM10 and NO2. PM10 and PM2.5 are strongly correlated (Pearson 0.72), PM10–NO2 moderately (0.39).

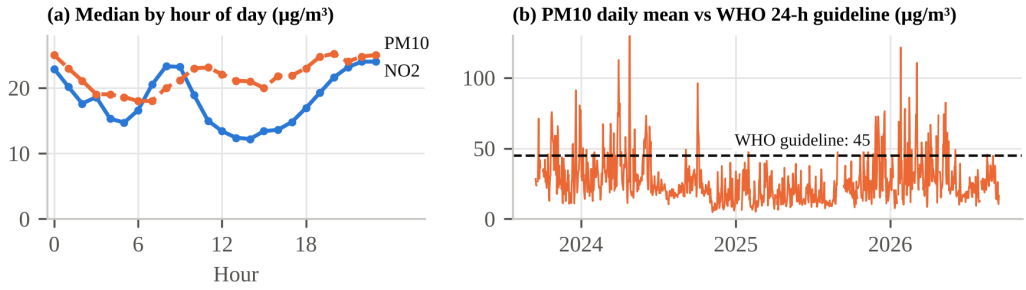


Figure 2. (a) Hourly medians of NO2 and PM10 show morning and evening peaks. (b) PM10 daily mean (days with ≥ 18 hourly values) against the WHO 24-hour guideline; gaps are days with insufficient data.

5.3 Modelling and First Experiments

Task: forecast the next day's daily median of each pollutant from the previous 7 days of all four pollutants (28 lag features). Daily rather than hourly values are used because hourly data are noisy and gappy, and daily values match how users plan and how WHO guidelines are defined. **Split:** chronological, without shuffling — ~60% training, ~15% validation (07.07–17.12.2025), 25% test (from 18.12.2025, kept unused for the final evaluation). **Models:** a persistence baseline ("tomorrow = today") and **Ridge regression** with standard scaling fitted on training data only; Ridge is simple, interpretable and handles strongly correlated lag inputs. Alpha was tuned over {0.1, …, 1000} on the validation period. **Metric:** MAE in µg/m³.

Table 1. Validation MAE (µg/m³), 07.07–17.12.2025; lower is better.

Model	PM10	PM2.5	SO2	NO2
Baseline ("tomorrow = today")	**7.57**	**2.63**	**1.19**	**9.58**
Ridge regression (best alpha)	7.67	3.40	1.51	11.42

First result: Ridge did not beat the baseline for any pollutant. Pollution is highly persistent from day to day, so persistence is a strong baseline. In addition, the model needed all 28 inputs, so only 48–49 of the 164 validation days could be evaluated. Planned for A2: per-pollutant models, fewer lags (e.g. 1, 2, 7 days), short-gap filling, weather features, Random Forest and XGBoost, and walk-forward validation.

5.4 Deliverables

A **Streamlit web app** with tomorrow's forecast, a WHO-based alert (computed from the daily mean, as the guidelines are defined), historical trends and SHAP explanations; the **GitHub repository** (data, notebooks, README, reproducible pipeline); Docker/cloud deployment; final report, presentation and demo video.

6. Limitations and Ethical Considerations

Single station (a pilot, not representative of all of Istanbul); no meteorological variables yet; unverified source time zone; "985" and PM2.5 > PM10 values unconfirmed without device information. The main ethical risk is a wrong "clean air" forecast for a sensitive user; we will show uncertainty, use cautious wording near thresholds, issue no forecast when inputs are missing, and state that the tool does not replace official warnings.

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References

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